

Daily Geometry Drill #3 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Relative Positions · Radical Axis by subtraction · Common Chords · Orthogonal Circles · Tangency to Both Axes.

Section A Rapid Recognition

1. The circle $x^2 + y^2 - 12x - 16y + 75 = 0$ is tangent to the circle $x^2 + y^2 = 25$. Write down the number of common tangents of the two circles.

Answer: 3

Method: $x^2 + y^2 - 12x - 16y + 75 = (x - 6)^2 + (y - 8)^2 - 100 + 75 = 0$, so centre (6, 8), radius $\sqrt{36 + 64 - 75} = 5$. Both circles have radius 5; $d = 10 = r_1 + r_2$ — externally tangent — 3 common tangents.

Investigate further: The bookkeeping: $d > r_1 + r_2 \Rightarrow 4$, $d = r_1 + r_2 \Rightarrow 3$, $|r_1 - r_2| < d < r_1 + r_2 \Rightarrow 2$, $d = |r_1 - r_2| \Rightarrow 1$, $d < |r_1 - r_2| \Rightarrow 0$. **“Tangent” means exactly one common point, giving exactly 3 tangents, not 2.**

2. The circles with centre (0,0) radius 5 and centre (13,0) radius 3 are disjoint. Write down the number of common tangents of the two circles.

Answer: 4

Method: $d = 13$, $r_1 + r_2 = 8$; since $d > r_1 + r_2$ the circles are fully disjoint and there are 4 common tangents.

Investigate further: Disjoint circles admit 2 direct (external) tangents plus 2 transverse (internal) tangents. **Once the circles overlap, the two transverse tangents disappear.**

3. Write down the radical axis of the circles $x^2 + y^2 - 6x + 2y + 1 = 0$ and $x^2 + y^2 - 2x - 4y + 13 = 0$.

Answer: $2x - 3y + 6 = 0$

Method: At the radical axis powers are equal, so subtract: $(-6x + 2y + 1) - (-2x - 4y + 13) = 0$, i.e. $-4x + 6y - 12 = 0$, i.e. $2x - 3y + 6 = 0$. The $x^2 + y^2$ terms cancel in one stroke.

Investigate further: Check the answer by symmetry: $x = \frac{3}{4}y - \frac{3}{2}$ plug-in balances both powers automatically. **A single-line subtraction beats solving any intersection; the radical axis exists even for disjoint circles.**

4. Write down whether the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 6x + 8y + 9 = 0$ are orthogonal (yes or no).

Answer: yes

Method: $x^2 + y^2 + 6x + 8y + 9$ has centre $(-3, -4)$ and $r^2 = 9 + 16 - 9 = 16$, so $r = 4$. $d = \sqrt{3^2 + 4^2} = 5$. Orthogonal iff $d^2 = r_1^2 + r_2^2$: $25 = 9 + 16$ ✓. Yes.

Investigate further: Equivalently with coefficients: $g_1 = f_1 = 0$, $g_2 = 3$, $f_2 = 4$ and $2g_1g_2 + 2f_1f_2 = 0$ must equal $c_1 + c_2 = (-9) + (9) = 0$. **Orthogonality is about the radii forming a right triangle with the centre distance.**

5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 10x + 5 = 0$ intersect in two points. Find the length of their common chord.

Answer: 8

Method: Subtract: $-25 = -10x + 5 \Rightarrow x = 3$, the radical axis. Distance of $x = 3$ from the origin is 3, radius 5, so half-chord $= \sqrt{25 - 9} = 4$, chord $= 8$.

Investigate further: The second circle has centre $(5, 0)$ and radius $\sqrt{20}$; the two circles agree on the two end-points of this chord. **Identify the radical axis first — the chord sits on it — then apply the half-chord formula.**

6. Two circles have radii 3 and 5 and their centres are 4 units apart. Write down the number of common tangents.

Answer: 2

Method: $|r_1 - r_2| = 2 < d = 4 < r_1 + r_2 = 8$ — intersecting circles — 2 common tangents.

Investigate further: Two crossing circles always have exactly 2 common tangents. **The trap is counting the two intersection points as “tangents”.**

7. The circle $x^2 + y^2 - 10x - 10y + k = 0$ is tangent to both coordinate axes. Find k .

Answer: 25

Method: Centre $(5, 5)$. Tangent to both axes forces the radius to equal each coordinate: $r = 5$. Then $r^2 = 25 = 5^2 + 5^2 - k$, so $k = 50 - 25 = 25$.

Investigate further: A circle tangent to both axes has centre (r, r) , $(-r, r)$, $(r, -r)$ or $(-r, -r)$ and equation $(x \mp r)^2 + (y \mp r)^2 = r^2$, constant r^2 . **The constant is r^2 , not $2r^2$.**

8. The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. Find C .

Answer: -11

Method: $g_1 = 2$, $f_1 = -1$; $g_2 = -2$, $f_2 = 2$, $c_2 = -1$. Orthogonality: $2g_1g_2 + 2f_1f_2 = c_1 + c_2 \Rightarrow 2(2)(-2) + 2(-1)(2) = C - 1 \Rightarrow -8 - 4 = C - 1 \Rightarrow C = -11$.

Investigate further: Geometric check: first circle centre $(-2, 1)$ with $r_1^2 = C$? No — $r_1^2 = 4 + 1 - C = 5 - C$; second centre $(2, -2)$ with $r_2^2 = 9$. $d^2 = 25 = 9 + 5 - C \Rightarrow C = -11$ ✓. **Both forms of the test mirror each other.**

9. How many common tangents do the circles $x^2 + y^2 - 4x - 6y + 12 = 0$ and $x^2 + y^2 + 2x - 10y + 25 = 0$ have?

Answer: 4

Method: First: centre $(2, 3)$, $r^2 = 4 + 9 - 12 = 1$. Second: centre $(-1, 5)$, $r^2 = 1 + 25 - 25 = 1$.

$d = \sqrt{(3)^2 + (2)^2} = \sqrt{13} > 2 = r_1 + r_2$ — disjoint unit circles — 4 common tangents.

Investigate further: Completing the square (Worksheet #2 skill) locates the centres; only then compare d with $r_1 + r_2$. **Worksheets #2's centre-and-radius reading is today's first move.**

10. Two circles have radii 7 and 3 and exactly 3 common tangents. Find the distance between their centres.

Answer: 10

Method: Exactly 3 common tangents means the circles are externally tangent, so $d = r_1 + r_2 = 7 + 3 = 10$.

Investigate further: Keep the five-case table on one line. **If there were 2 tangents you would instead need $|3 - 7| < d < 10$, which leaves a whole range, not a single value.**

Section B Manipulation Drills

1. How many common tangents do the circles $x^2 + y^2 = 9$ and $x^2 + y^2 - 4x - 6y + 9 = 0$ have?

- A) 0
- B) 1
- C) 2
- D) 3
- E) 4

Answer: C) 2

Method: $x^2 + y^2 - 4x - 6y + 9 = 0$: centre $(2, 3)$, $r^2 = 4 + 9 - 9 = 4$. The other has centre $(0, 0)$ radius 3. $d = \sqrt{13}$; $|3 - 2| = 1 < \sqrt{13} < 5$ — intersecting — 2 common tangents.

Investigate further: $x^2 + y^2 = 9$ is radius 3, so the second circle is strictly inside the first's 5-radius relation. **Do not stop after finding the centres; compare d against both $r_1 + r_2$ and $|r_1 - r_2|$.**

2. The radical axis of the circles $x^2 + y^2 + ax + by + c = 0$ and $x^2 + y^2 + 6x - 4y + 10 = 0$ is $2x - y + 3 = 0$. What is $a + b$?

- A) 3
- B) -3
- C) 13
- D) -13

Answer: A) 3

Method: Subtract: $(a - 6)x + (b + 4)y + (c - 10) = 0$. Compare with $2x - y + 3 = 0$: $a - 6 = 2 \Rightarrow a = 8$; $b + 4 = -1 \Rightarrow b = -5$; $c - 10 = 3 \Rightarrow c = 13$. So $a + b = 3$.

Investigate further: Each coefficient matches with the *same* scale factor (here 1); change the factor and all three coefficient equations change together. **The y -coefficient sign is the usual slip — $b + 4 = -1$ gives $b = -5$, not -3 .**

3. The circle $x^2 + y^2 - 4x + 2y - 5 = 0$ is orthogonal to $x^2 + y^2 + 6x - 2y + k = 0$. Find k .

- A) -9

- B) 9
C) -5
D) 14

Answer: A) -9

Method: $g_1 = -2, f_1 = 1, c_1 = -5; g_2 = 3, f_2 = -1, c_2 = k$. $2g_1g_2 + 2f_1f_2 = 2(-2)(3) + 2(1)(-1) = -14 = c_1 + c_2 = -5 + k \Rightarrow k = -9$.

Investigate further: Orthogonal circles cut at right angles: the radii to a common point are perpendicular. **Never pair g with c or f with g — the pairing is $g_1g_2 + f_1f_2$ against $\frac{c_1+c_2}{2}$.**

4. A circle is tangent to both coordinate axes in the first quadrant and passes through the point (1, 8). Find the smaller possible radius.

Answer: 5

Method: Centre (r, r) , radius r : $(1 - r)^2 + (8 - r)^2 = r^2 \Rightarrow r^2 - 18r + 65 = 0 \Rightarrow (r - 5)(r - 13) = 0$. Smaller radius 5.

Investigate further: There really are two such circles — always state which root the question wants. **Both roots are positive, so the phrasing “smaller” is essential.**

5. Two circles of equal radius 5 have centres (0, 0) and (6, 0). Their radical axis has equation?

- A) $x = 3$
B) $x = 6$
C) $y = 3$
D) $y = 0$

Answer: A) $x = 3$

Method: Equal radii: subtract $(x^2 + y^2 - 25) = ((x - 6)^2 + y^2 - 25) \Rightarrow -25 = -12x + 36 - 25 \Rightarrow x = 3$.

Investigate further: For equal circles the radical axis is the perpendicular bisector of the centre segment. **A quick sanity check: power at (3, 0) is $9 - 25 = -16$ for both sides.**

6. Two circles of radii 4 and 6 have centres 9 units apart. How many common tangents do they have?

- A) 0
B) 1
C) 2
D) 3
E) 4

Answer: C) 2

Method: $|r_1 - r_2| = 2 < d = 9 < 10 = r_1 + r_2$: strict overlap without containment — 2 common tangents.

Investigate further: Compare d to 10 (sum) and 2 (difference) before counting. 9 is **dangerously close to 10; only exactly 10 would give 3.**

7. The circle $x^2 + y^2 = 4$ is orthogonal to $x^2 + y^2 + 8x - 6y + C = 0$. Find C .

- A) 4
B) -4
C) 25
D) 2

Answer: A) 4

Method: Second circle: centre $(-4, 3)$, $r_2^2 = 25 - C$. $d^2 = 16 + 9 = 25$. Orthogonal: $d^2 = r_1^2 + r_2^2 = 4 + (25 - C) = 29 - C \Rightarrow C = 4$.

Investigate further: Check with $g_1 = f_1 = 0$: the coefficient test gives $0 = -4 + C \Rightarrow C = 4 \checkmark$.
Both routes agree — use whichever you trust.

8. The circles $x^2 + y^2 = 36$ and $(x - 6)^2 + y^2 = 36$ intersect. Find the length of their common chord.

Answer: $6\sqrt{3}$

Method: Equal radii 6 on centres 6 apart: radical axis is the perpendicular bisector $x = 3$. Half-chord $= \sqrt{36 - 9} = 3\sqrt{3}$, chord $= 6\sqrt{3}$.

Investigate further: The two circles overlap in a lens whose extreme width is $6\sqrt{3}$. **Do not add the two radii; the chord lives on the radical axis, not at the intersection of centre-lines.**

9. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(4, 8)$. Its possible radii are?

- A) 4 only
B) 20 only
C) 4 and 20
D) 16

Answer: C) 4 and 20

Method: Centre (r, r) : $(4 - r)^2 + (8 - r)^2 = r^2 \Rightarrow r^2 - 24r + 80 = 0 \Rightarrow (r - 4)(r - 20) = 0$. Both 4 and 20.

Investigate further: Verify: radius 4 circle touches both axes near the origin; radius 20 circle is far out and still reaches $(4, 8)$. **A quadratic in r can give two genuine circles — list both.**

10. The point $P = (1, t)$ lies on the radical axis $3x - 4y + 5 = 0$ of two circles. Find t .

Answer: 2

Method: Substitute: $3(1) - 4t + 5 = 0 \Rightarrow 8 - 4t = 0 \Rightarrow t = 2$.

Investigate further: The radical axis is exactly the set of points of equal power — a line. **Put P into the line, not into either circle equation.**

Section C Substitution & Structure

1. Two circles have the same radius. Their centres are $C_1(-2, 1)$ and $C_2(3, -2)$. The circles intersect in two distinct points P and Q . Which is the equation of the line PQ ? (after TMUA 2021 Paper 1 Q1)

- A) $5x + 3y = 4$
 B) $3x - 5y = 4$
 C) $5x - 3y = 4$
 D) $5x - 3y = 1$

Answer: C) $5x - 3y = 4$

Method: Equal radii: subtract $(x+2)^2 + (y-1)^2 = (x-3)^2 + (y+2)^2$, i.e. $x^2 + 4x + 4 + y^2 - 2y + 1 = x^2 - 6x + 9 + y^2 + 4y + 4$, so $10x - 6y = 8$, i.e. $5x - 3y = 4$. The $x^2 + y^2$ cancel in one line. The midpoint $(\frac{1}{2}, -\frac{1}{2})$ satisfies $5(\frac{1}{2}) - 3(-\frac{1}{2}) = 4$ ✓.

Investigate further: Equal radii make the radical axis the perpendicular bisector of C_1C_2 , but you still get it fastest by subtraction. **Option A flips the sign of y ; option D is a parallel line through the origin.**

2. The circles $C_1 : (x-1)^2 + y^2 = 4$ and $C_2 : (x-5)^2 + (y-3)^2 = 9$ have radii 2 and 3. How many common tangents do C_1 and C_2 have? (after TMUA 2019 Paper 1 Q7)

- A) 0
 B) 1
 C) 2
 D) 3
 E) 4

Answer: D) 3

Method: $C_1 = (1, 0)$, $C_2 = (5, 3)$, $d = \sqrt{4^2 + 3^2} = 5$. $r_1 + r_2 = 5$, so $d = r_1 + r_2$ — externally tangent — exactly 3 common tangents (2 direct plus the common tangent at the touch point).

Investigate further: $d > r_1 + r_2 \Rightarrow 4$, $d = r_1 + r_2 \Rightarrow 3$, $|r_1 - r_2| < d < r_1 + r_2 \Rightarrow 2$, $d = |r_1 - r_2| \Rightarrow 1$, $d < |r_1 - r_2| \Rightarrow 0$. **The 3-4-5 triple makes d instant.**

3. The circles $x^2 + y^2 = 169$ and $x^2 + y^2 - 6x + 8y - 119 = 0$ meet at P and Q . What is the length PQ ? (after TMUA 2016 Paper 2 Q9)

- A) 12
 B) 16
 C) 24
 D) 26

Answer: C) 24

Method: Second circle: centre $(3, -4)$, $r^2 = 9 + 16 + 119 = 144$, $r = 12$. Subtract: $-169 = -6x + 8y - 119 \Rightarrow 6x - 8y - 50 = 0 \Rightarrow 3x - 4y - 25 = 0$. Distance from origin $= |-25|/5 = 5$. Half-chord

$= \sqrt{169 - 25} = 12$, chord = 24. Check from $(3, -4)$: distance to line $|9 - 16 - 25|/5 = 1$, half-chord $\sqrt{144 - 1} = \sqrt{143}$. wait $r = 12$, $d = 1$ from second centre? No $d = |9 + 16 - 25|/5 = 0$? Let's recompute: $3(3) - 4(-4) - 25 = 9 + 16 - 25 = 0$? Actually $9 + 16 - 25 = 0$, so second centre lies on radical axis? Check again: line $3x - 4y - 25 = 0$ through $(3, -4)$ indeed $9 + 16 - 25 = 0$. So chord is diameter of second circle? No, check $r_2 = 12$, $r_1 = 13$, $d_{centres} = 5$, not on axis? Wait calculation shows $3x - 4y = 25$ passes through $(3, -4)$? $9 + 16 = 25$ yes. That suggests second centre is on radical axis, which cannot be unless equal radii? Something off: original circle $x^2 + y^2 - 6x + 8y - 119 = 0$ centre $(3, -4)$ radius $\sqrt{9 + 16 + 119} = 12$, first circle origin radius 13. Radical axis is $6x - 8y = 50$, i.e. $3x - 4y = 25$. Plug $(3, -4)$: $9 + 16 = 25$ true, so second centre lies on axis — that would mean power of second centre is same? But power of centre w.r.t own circle is $-r^2 = -144$, w.r.t first circle is $9 + 16 - 169 = -144$ same, so indeed second centre has equal power because $d^2 = 25$, $r_1^2 - r_2^2 = 169 - 144 = 25$, so radical axis passes through second centre's projection? Actually condition $d^2 = r_1^2 - r_2^2$ makes axis pass through second centre, possible when $d^2 = r_1^2 - r_2^2$ gives one centre on axis. Here $25 = 25$ holds, so yes axis passes through $(3, -4)$. Then distance from second centre to axis is 0, chord would be diameter 24 of second circle, consistent! So half-chord from origin 12 matches $r_2 = 12$ diameter. Both views agree: chord = 24.

Investigate further: The 5-12-13 triple runs the whole question. Option D 26 is the diameter of the large circle, strictly longer than any chord. **The radical axis can pass through a centre when $d^2 = |r_1^2 - r_2^2|$.**

4. A circle in the first quadrant is tangent to both coordinate axes and passes through the point $(9, 2)$. Its two possible radii are $r_1 < r_2$. What is $r_1 + r_2$? (*adapted from JZMaths Mock 1 Q5*)

- A) 14
B) 17
C) 22
D) 34

Answer: C) 22

Method: Centre (r, r) , radius r : $(9 - r)^2 + (2 - r)^2 = r^2 \Rightarrow r^2 - 22r + 85 = 0$. By Vieta sum = 22, product = 85; roots are 5 and 17. Sum 22.

Investigate further: Both $r = 5$ and $r = 17$ give genuine circles. **Option B 17 is the larger root alone; option A 14 would be 9 + 5 misreading.**

5. The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. What is C ? (*after Oxbridge Paper 1 Q8*)

- A) -11
B) -8
C) 11
D) 5

Answer: A) -11

Method: $g_1 = 2$, $f_1 = -1$; $g_2 = -2$, $f_2 = 2$, $c_2 = -1$. Orthogonality $2g_1g_2 + 2f_1f_2 = c_1 + c_2 \Rightarrow 2(2)(-2) + 2(-1)(2) = C - 1 \Rightarrow -8 - 4 = C - 1 \Rightarrow C = -11$. Geometric check: centres $(-2, 1)$ and $(2, -2)$, $d^2 = 25$, $r_1^2 = 5 - C$, $r_2^2 = 9$, $25 = 5 - C + 9 \Rightarrow C = -11 \checkmark$.

Investigate further: Pair g with g , f with f : $2g_1g_2 + 2f_1f_2$ against $(c_1 + c_2)$. **Option C** 11 flips the sign of the constant.

6. A circle has equation $x^2 + ax + y^2 + by + c = 0$ where $a, b, c \neq 0$. Which condition is necessary and sufficient for the circle to be tangent to the y -axis? (after TMUA 2021 Paper 2 Q15)

- A) $a^2 = 4c$
 B) $b^2 = 4c$
 C) $a^2 + b^2 = 4c$
 D) $c = 0$

Answer: B) $b^2 = 4c$

Method: Centre $(-a/2, -b/2)$, $r^2 = a^2/4 + b^2/4 - c$. Tangent to y -axis ($x = 0$) needs distance $|-a/2| = r$, so $a^2/4 = a^2/4 + b^2/4 - c \Rightarrow b^2 = 4c$. (Tangency to x -axis would be $a^2 = 4c$.)

Investigate further: The y -axis condition kills b , not a , because the radius condition cancels $a^2/4$. **Option A** $a^2 = 4c$ is tangency to the x -axis; **option D** $c = 0$ is through the origin.

7. The equal circles $C_1 : (x + 2)^2 + (y - 1)^2 = 3$ and $C_2 : (x - 4)^2 + (y - 1)^2 = 3$ have centres 6 apart. A transverse common tangent with positive gradient makes an acute angle θ with the x -axis. What is $\tan \theta$? (adapted from TMUA 2020 Paper 1 Q16)

- A) $\frac{1}{2}$
 B) $\frac{\sqrt{2}}{2}$
 C) $\sqrt{2}$
 D) $\frac{\sqrt{3}}{3}$

Answer: B) $\frac{\sqrt{2}}{2}$

Method: Equal radii $\sqrt{3}$ at $(-2, 1)$ and $(4, 1)$. Transverse tangent passes through midpoint $(1, 1)$: line $y - 1 = m(x - 1)$, i.e. $mx - y + 1 - m = 0$. Distance from $(-2, 1)$ to line $= |-3m|/\sqrt{m^2 + 1} = \sqrt{3} \Rightarrow 9m^2 = 3(m^2 + 1) \Rightarrow 6m^2 = 3 \Rightarrow m^2 = 1/2 \Rightarrow m = \sqrt{2}/2$ (positive).

Investigate further: The transverse tangent crosses the centre line at the midpoint; the direct common tangents are horizontal ($m = 0$). **Option C** $\sqrt{2}$ inverts the fraction; **option A** $1/2$ squares the root.

8. The three circles $C_1 : x^2 + y^2 + 4x - 1 = 0$, $C_2 : x^2 + y^2 - 8y + 11 = 0$, $C_3 : x^2 + y^2 + 2x - 12y + 13 = 0$ have pairwise radical axes concurrent at R . What are the coordinates of R ? (adapted from Vantage Mock Paper 2 Q10 / BMO1 2017 Q4)

- A) $(1, 1)$
 B) $(1, -1)$
 C) $(3, 1)$
 D) $(1, 2)$

Answer: A) $(1, 1)$

Method: Radical axis $C_1 - C_2$: $(4x - 1) - (-8y + 11) = 0 \Rightarrow 4x + 8y - 12 = 0 \Rightarrow x + 2y - 3 = 0$.
 $C_2 - C_3$: $(-8y + 11) - (2x - 12y + 13) = 0 \Rightarrow -2x + 4y - 2 = 0 \Rightarrow x - 2y + 1 = 0$. Solve $x + 2y = 3$,
 $x - 2y = -1 \Rightarrow x = 1, y = 1$. Power: $1 + 1 + 4 - 1 = 5$ for all three $\checkmark$.

Investigate further: Any two radical axes intersect at the radical centre; the third must pass through it. **Option D** $(1, 2)$ lies on only one axis.

Section D Challenge Ramp

1. The circles $C_1 : (x + 2)^2 + (y - 3)^2 = 18$ and $C_2 : (x - 7)^2 + (y + 6)^2 = 2$ have centres O_1 and O_2 . What is the shortest distance between a point on C_1 and a point on C_2 ? (*adapted from TMUA 2018 Paper 1 Q3*)

A) $5\sqrt{2} - 4$

B) $5\sqrt{2} - 5$

C) $5\sqrt{2}$

D) $9\sqrt{2}$

Answer: C) $5\sqrt{2}$

Method: $O_1 = (-2, 3)$, $O_2 = (7, -6)$, $O_1O_2 = \sqrt{9^2 + 9^2} = 9\sqrt{2}$, $r_1 = \sqrt{18} = 3\sqrt{2}$, $r_2 = \sqrt{2}$, sum $4\sqrt{2}$, shortest $= 9\sqrt{2} - 4\sqrt{2} = 5\sqrt{2}$. Externally disjoint ($d > r_1 + r_2$).

Investigate further: Shortest $= d - r_1 - r_2$ when disjoint; if one contained the other it would be $|r_1 - r_2| - d$. **Option D** $9\sqrt{2}$ is O_1O_2 itself, forgetting to subtract radii.

2. The circles $(x + 4)^2 + (y + 1)^2 = 64$ and $(x - 8)^2 + (y - 4)^2 = r^2$ ($r > 0$) have exactly one point in common. What is the difference between the two possible values of r ? (*after TMUA 2019 Paper 1 Q6*)

A) 8

B) 16

C) 26

D) 42

Answer: B) 16

Method: Centre distance $d = \sqrt{12^2 + 5^2} = 13$, $r_1 = 8$. One point $\Rightarrow |r - 8| = 13$ (internal) or $r + 8 = 13$ (external) $\Rightarrow r = 21$ or 5 , difference 16.

Investigate further: External tangency gives smaller $r = 5$, internal gives larger 21. **Option A** 8 is r_1 itself; **option D** 42 would be sum not difference.

3. Three circles of radii 3, 4 and 5 are pairwise externally tangent. What is the area of the triangle formed by joining their centres? (*adapted from BMO1 2017 Q4*)

A) $12\sqrt{5}$

B) 12

C) 84

D) 30

Answer: A) $12\sqrt{5}$

Method: Centre triangle sides $3 + 4 = 7$, $3 + 5 = 8$, $4 + 5 = 9$. $s = 12$, Heron area $= \sqrt{12 \cdot 5 \cdot 4 \cdot 3} = \sqrt{720} = 12\sqrt{5}$. Coordinate check: $(0, 0)$, $(7, 0)$, third $x = 33/7$, $y = 24\sqrt{5}/7$, area $= \frac{1}{2} \cdot 7 \cdot 24\sqrt{5}/7 = 12\sqrt{5}$ ✓.

Investigate further: Side is sum of radii, not difference. **Option C** 84 is area of 13-14-15 triangle, the classic Heron triple.

4. Two circles of radii 3 and 4 are orthogonal. What is the length of their common chord? (after TMUA 2016 Paper 2 Q9 / Oxbridge)

A) $\frac{24}{5}$ B) $\frac{12}{5}$

C) 6

D) $\frac{48}{5}$ **Answer:** A) $\frac{24}{5}$

Method: $d^2 = r_1^2 + r_2^2 = 9 + 16 = 25$, $d = 5$. Place centres $(0, 0)$ and $(5, 0)$; radical axis $10x = 18 \Rightarrow x = 9/5$ from small centre. Half-chord $\sqrt{9 - (9/5)^2} = \sqrt{144/25} = 12/5$, chord $= 24/5$. Also chord $= 2r_1r_2/d = 24/5$.

Investigate further: Option C 6 is diameter of small circle; chord is strictly shorter. **Option B** $12/5$ is half the chord.

5. A point P has equal power with respect to the circles $x^2 + y^2 = 1$ and $(x - 6)^2 + y^2 = 4$. What does the locus of P describe? (adapted from JZMaths / Vantage)

A) the line $x = \frac{11}{4}$ B) the circle $x^2 + y^2 = 1$ C) the y -axisD) a single point $\left(\frac{11}{4}, 0\right)$ **Answer:** A) the line $x = \frac{11}{4}$

Method: Equal powers = radical axis: $x^2 + y^2 - 1 = (x - 6)^2 + y^2 - 4 = x^2 - 12x + 32 + y^2 \Rightarrow 12x = 33 \Rightarrow x = 11/4$, a vertical line. Exists even though circles are disjoint ($d = 6 > 3$).

Investigate further: Locus of equal power is always a straight line, even for disjoint circles. **Option D** is a single point on that line, not the whole locus.

Top 5 Patterns — Worksheet #3

- 1. Classify two circles by comparing d with $r_1 + r_2$ and $|r_1 - r_2|$.** $d > r_1 + r_2$ (4 tangents), $d = r_1 + r_2$ (3), overlap (2), $d = |r_1 - r_2|$ (1), containment (0). (see A1, A2, A6, B1, B6, C2, D1, D2)
- 2. Radical axis = the two equations subtracted.** The $x^2 + y^2$ terms vanish in one line; the result is the common chord for intersecting circles and the equal-power line in general. (see A3, B2, C1, C3, C8, D5)
- 3. Common chord = radical axis + half-chord formula.** Find the radical axis, take its distance d from a centre, then $2\sqrt{r^2 - d^2}$. (see A5, B8, C3, D4)
- 4. Orthogonality either way:** $d^2 = r_1^2 + r_2^2$ or $2g_1g_2 + 2f_1f_2 = c_1 + c_2$. The radii make a right triangle against the centre distance. (see A4, A8, B3, B7, C5, D4)
- 5. A circle tangent to both axes lives at $(\pm r, \pm r)$ with constant r^2 .** Powers of r are found from a point condition; Vieta gives sum and product instantly. (see A7, B4, B9, C4)

Common Traps — Worksheet #3

- 1. Miscounting the common-tangent table.** Externally tangent means exactly 3 common tangents (diameter line plus two direct); crossing circles have 2, disjoint circles 4. One boundary-off-by-one changes the whole count. (see A1, A2, A6, A10, B1, B6, C2, D1, D2)
- 2. Mangling the subtraction sign on the radical axis.** It is the difference of the two *constants and linear terms* set to zero; flip a sign and you get the next option in C1 or C5. (see A3, B2, C1, C5, C8)
- 3. Wrong pairing in the orthogonality test.** The condition binds $g_1g_2 + f_1f_2$ against $(c_1 + c_2)/2$ — pair g with g , f with f . (see A8, B3, B7, C5, D4)
- 4. Imagining the radical axis is a circle.** Equal powers trace a straight line, not a curve, even between disjoint circles. (see B5, B10, C1, D5)
- 5. Forgetting the triangle of centres on tangent circles.** For pairwise tangent circles each centre distance is the *sum* of the two radii; subtracting instead of adding breaks Heron immediately. (see D3, C2, A10)

“Two circles are solved by one subtraction: locate, subtract, then compare with the sum and difference of the radii.”

Found a mistake or want to contribute a sheet?
 Visit the open-source project at github.com/The-CerealDev/speed-maths